IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART II: MEng, BEng and ACGI

POWER ENGINEERING

Monday, 1 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T.C. Green
Second Marker(s) : B. Chaudhuri

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Information for Candidates

Electronic charge: $e = 1.60 \times 10^{-19}$ C

Boltzmann constant: $k = 1.38 \times 10^{-23}$ J/K

Permittivity of free space: $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

Permeability of free space: $\mu_0 = 4\pi \times 10^{-7}$ H/m

Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_O}{V_I} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1+\frac{2fL\,I_O}{V_I\delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_O}{V_I} = \frac{1}{1-\delta}$ or $\frac{V_O}{V_I} = \frac{1}{1-\frac{V_I\delta^2}{2fL\,I_I}}$

Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}\,V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\,I_P$

Power $P_{3\phi} = 3\,V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Induction Machines

$\omega_S = {\omega_E}/{P}$; s $= \frac{\omega_S - \omega_R}{\omega_S}$; $T_{EM} = \frac{3\,I'^2_R\,R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}} cos(\theta) - \frac{V_R V_S}{Z_{SR}} cos(\theta + \delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}} sin(\theta) - \frac{V_R V_S}{Z_{SR}} sin(\theta + \delta)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\,P_S + X\,Q_S}{|V_S|}$

Photovoltaic Systems

$$I_{PV} = I_{Ph} - I_{AK} - I_{Sh} \qquad I_{AK} = I_0\left(e^{\frac{V_{PV} + I_{PV}R_S}{K_I\,v_T}} - 1\right)$$

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1. This question covers several topics and all parts should be attempted.

a) An electricity grid is divided into a transmission grid and a distribution grid.

i) Explain their different roles. [4]

ii) Explain why a transmission system is constructed with meshed double circuits and a distribution system with radial single circuits. [4]

b) Consider an overhead line of 3 km used for a single-phase AC supply of 11 kV at 50 Hz and with two conductors of aluminium (conductivity of 35×10^6 S/m) each with a radius of 5.0 mm. The conductors are spaced 1.0 m apart on the cross-arms of the towers.

i) Explain why one expects an overhead line to have a low capacitance per unit length and a high inductance per unit length compared to a cable of equivalent voltage and current rating. [4]

ii) Calculate the inductive reactance and resistance of the route (using the approximate form of the equations). [4]

iii) Estimate the voltage magnitude difference between sending and receiving ends of the overhead line when real power of 1 MW is consumed at the receiving end with a power factor of 0.95 lagging. [4]

c) Consider a three-phase AC supply with a line voltage of 400 V.

i) Three impedances of $(30 - j\,12)\ \Omega$ are formed into a delta connection and connected to the supply. Calculate the phase current and line currents that flow. Provide magnitudes of both currents and angle of a phase current relative to a phase voltage at 0°. [4]

ii) Calculate the real and reactive powers drawn by the delta-connected impedances. [4]

d) Consider a typical photovoltaic (PV) panel.

i) Explain why not all photons of light that impinge on the panel are captured and converted to usable electrical energy. [4]

ii) It is proposed to increase the efficiency of a PV panel by creating two *p*-*n* junctions, one in front of the other, with different band-gaps. Explain why this might be thought to increase efficiency **and** explain whether the low or high band-gap junction should be placed first in the path of the incident light. [4]

iii) A particular photovoltaic panel produces a short-circuit current of 2.5 A under an irradiance of 1,000 W/m^2. The reverse saturation current of the panel was measured as 2×10^{-10} A at 30°C. Calculate the open circuit voltage under the same irradiance and at a temperature of 30°C. You may assume that the shunt resistance of the panel is negligible and that the ideality factor is 1. [4]

2. A **boost** switch-mode power supply (SMPS) is to be designed for an input voltage of 10 V and output voltage of 30 V. The SMPS will be required to process a maximum throughput power of 40 W and have an output voltage ripple which is no greater than 75 mV.

The SMPS will be designed to operate in continuous conduction mode down to a throughput power of 0.5 W.

The on-state resistance of the chosen MOSFET is 0.05 Ω. The turn-on and turn-off energy losses of the MOSFET are 2 μJ and 3 μJ when operated at 1 A and scale linearly with current.

i) Calculate the value of the input current when the circuit operates at maximum power. [2]

ii) Calculate the value of duty-cycle at which the SMPS should be operated under continuous conduction. [2]

iii) Sketch graphs of the current through the inductor, the diode and the capacitor for operation at maximum power. [4]

iv) Calculate the switching frequency that should be chosen to keep the power dissipation in the MOSFET to a maximum of 1 W. [4]

v) It is required that the circuit operate in continuous mode down to a throughput power of 0.5 W. Calculate the value of inductor that should be chosen to achieve this. [6]

vi) It is intended to use an output-side capacitor of 470 μF. Calculate the maximum value of effective series resistance (ESR) that this capacitor could be allowed to have. You may approximate where needed. [8]

vii) This SMPS will be used with a feedback control system that will set duty-cycle to achieve the desired output voltage. Consider the case of a throughput of 0.25 W. Calculate the duty-cycle the controller would need to apply to maintain the output voltage at 30 V. [4]

3. An induction machine which is a 3-phase, 2 pole-pair, star-connected machine is connected to a 50 Hz supply with a phase-voltage of 230 V. The parameters of the equivalent circuit, referred to the stator, are:

Stator resistance,	R_S = 1.4 Ω.
Stator leakage reactance,	X_S = 2.0 Ω;
Magnetising resistance,	R_M = 200 Ω;
Magnetising reactance,	X_M = 40 Ω;
Referred rotor leakage reactance,	X_R = 1.5 Ω;
Referred rotor resistance,	R_R = 0.7 Ω;

The machine is operated with a slip of s = 0.02.

i) Give an expression for the total input impedance of the machine and calculate its value. [8]

ii) Calculate the stator current and referred rotor current. [5]

iii) Calculate the electro-magnetic torque developed. [5]

iv) Re-calculate the electro-magnetic torque developed for the case of s = 0.04 [5]

v) Explain, with reference to the induced voltage in the rotor windings, why the torque is expected to be an approximately linear function of slip for small slip (operation close to synchronous speed) **and** assess the linearity by comparing the answers in parts (iii) and (iv). [7]